METRO TICKET GENERATING SYSTEM

Project Design Phase-II – Data Flow Diagram & User Stories

Date	**2 October 2026**
Team ID	
Project Name	Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows
Maximum Marks	**4 Marks**

Data Flow Diagram Description

The Data Flow Diagram represents how a metro ticket booking or smart card recharge request moves through the ServiceNow solution. A commuter submits the Metro Ticket Request catalog item, the request is stored as a Requested Item (RITM), dynamic distance- and zone-based fare calculations are computed via backend scripts, a unique digital QR-code payload is generated upon approval/validation, and a fulfillment task is routed to the station operator workspace team alongside automated confirmation email notifications.

User Stories

- **USN-1 (Commuter / Epic: Metro Ticket Request):** As a commuter, I can submit a metro ticket request by selecting my origin and destination stations. *Acceptance Criteria:* The request is submitted and a Requested Item (RITM) is created. *Priority:* High *Release:* Sprint-1
- **USN-2 (Commuter / Epic: Metro Ticket Request):** As a commuter, I can select the ticket type, travel class, and smart card number. *Acceptance Criteria:* The entered values are saved with the request. *Priority:* High *Release:* Sprint-1
- **USN-3 (Commuter / Epic: Dynamic Fare Calculation):** As a commuter, I want the system to automatically compute the correct fare based on station zones. *Acceptance Criteria:* The fare is calculated dynamically via script and stored with the booking record. *Priority:* High *Release:* Sprint-1
- **USN-4 (Commuter / Epic: Digital QR Ticket Generation):** As a commuter, I want a unique digital QR-code ticket generated upon booking confirmation. *Acceptance Criteria:* A scannable QR payload is linked to my booking record and visible on the portal. *Priority:* High *Release:* Sprint-2
- **USN-5 (Station Operator / Epic: Operator Task & Validation):** As a station operator, I can receive fulfillment tasks and validate active QR passes via the workspace. *Acceptance Criteria:* A catalog task is created and assigned to the operator queue for verification. *Priority:* High *Release:* Sprint-2
- **USN-6 (Commuter / Epic: Notification):** As a commuter, I can receive an email notification after my ticket request is confirmed. *Acceptance Criteria:* The configured email notification is generated successfully. *Priority:* Medium *Release:* Sprint-2
- **USN-7 (Administrator / Epic: Request Monitoring):** As an administrator, I can monitor requests, dynamic pricing scripts, and operator tasks. *Acceptance Criteria:* The related ServiceNow records and flow execution can be reviewed. *Priority:* Medium *Release:* Sprint-2